

Milestone 5: Simulation Runs and Data Collection

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Abstract—This milestone presents a comprehensive soccer match simulation system for UEFA Champions League matches. The system employs discrete-event simulation and machine learning to model match dynamics at a second-by-second resolution. Two complementary models form the core of the system: a Large Events Model (LEM) that predicts on-ball events and a transformer-based Off-Ball Movement Predictor that models player positioning. The simulation captures player attributes, team tactics, and match context to generate realistic match flows. This report details the simulation architecture, execution methodology, data collection processes, and analysis of simulation runs. Results demonstrate that the integration of discrete-event simulation with machine learning effectively captures fundamental soccer match dynamics, though certain aspects require further refinement. Based on these findings, we propose several enhancements for future development, including improved contextual awareness, enhanced data integration, system optimization, and validation framework improvements.

Index Terms—soccer simulation, discrete-event simulation, machine learning, sports analytics, event prediction, player movement, tactical analysis

I. INTRODUCTION

A. Project Overview

This project creates a comprehensive simulation of UEFA Champions League (UCL) matches using discrete-event simulation and machine learning approaches. The simulation models individual player actions, team tactics, and match dynamics at a second-by-second resolution, capturing the complexity and fluidity of professional football. By integrating detailed player statistics, team formations, and tactical styles, the simulation provides a realistic representation of how matches unfold, enabling analysis of various scenarios and outcomes.

B. Purpose of Simulation Runs

The simulation runs serve multiple essential purposes for this project:

- To validate the accuracy and realism of our discrete-event simulation approach
- To generate sufficient data for evaluating the performance of our machine learning models
- To identify areas for improvement in both simulation logic and ML model training
- To establish benchmarks for future enhancements and refinements
- To demonstrate the capability of the simulation to capture key dynamics of football matches

C. Key Questions to Address

- 1) How accurately does our simulation capture the flow and statistical patterns of real UEFA Champions League matches?
- 2) What insights can be derived about the relationship between player attributes, team tactics, and match outcomes?
- 3) How effectively do our machine learning models predict player actions and movements?
- 4) What are the current limitations of our simulation approach, and how might they be addressed?
- 5) How can the simulation be optimized for both computational efficiency and realistic representation?

II. SIMULATION OUTLINE

A. Simulation Description

Our simulation employs a discrete-event framework to model UCL matches with high fidelity. At its core are two complementary machine learning components:

- 1) **Large Events Model (LEM)** - Inspired by the approach in the `nvsclub/LargeEventsModel` repository, this component predicts on-ball events such as passes, shots, dribbles, and tackles. It takes the current match state as input and outputs probabilities for various possible actions, their spatial parameters, and success likelihoods.
- 2) **Off-Ball Movement Predictor** - A transformer-based architecture that models how players without the ball position themselves based on game state, team tactics, and individual attributes. This component is crucial for capturing the emergent spatial patterns that characterize different playing styles.

The simulation progresses in discrete time steps (generally one second), with events scheduled and processed sequentially. Each event triggers updates to the match state, including player positions, ball location, possession, and score. The system maintains a comprehensive record of events and player movements throughout the match.

Key components of the simulation include:

- **Event Generation and Processing:** Events such as passes, shots, and tackles are predicted and processed
- **Player Movement:** Both on-ball and off-ball movements are modeled
- **Physical Parameters:** Player attributes like pace, strength, and stamina affect action outcomes
- **Tactical Implementation:** Team formations and styles influence player positioning and decision-making

- **Probabilistic Outcomes:** Event success is determined probabilistically based on player skills, match context, and opposing players

B. Execution Instructions

To execute the simulation, follow these steps:

1) Data Preparation:

- Ensure that player data (`players.csv`), team data (`teams.csv`), and team style data (`team_styles.csv`) are available in the `csv_exports` directory
- If using pre-trained models, ensure they are located in the `models` directory

2) Running a Basic Simulation:

This runs a 90-minute match simulation between teams with IDs 57 and 39, with visualization enabled.

3) Running a Full Simulation:

```
python simulation_engine.py
```

This executes a complete match simulation using the `SoccerMatchSimulator` class, with advanced event processing and player movement.

4) Output:

- Events are recorded in a structured format (CSV/-database)
- Visualizations show player positions and movements
- Match statistics are calculated and displayed
- For detailed analysis, the simulation can export event and position data

C. Assumptions and Limitations

The current simulation operates under several assumptions and has certain limitations:

1) Assumptions:

- Player performance is consistent throughout the match, with only gradual fatigue effects
- Team tactics remain generally constant, with no major strategic shifts mid-match
- All players behave rationally according to their tactical roles and game context
- Environmental factors (weather, pitch conditions) are not considered
- Psychological factors (pressure, momentum) are minimally modeled

2) Limitations:

- The simulation does not yet account for complex tactical adaptations by managers
- Injury mechanics are simplified and may not accurately reflect real-world patterns
- Referee decisions and set-piece tactics have limited sophistication
- The current model lacks sufficient training data for certain rare match scenarios
- Computational constraints limit the complexity of real-time decision-making

- Player interactions beyond basic positional relationships are not fully modeled
- The simulation does not yet capture emotional or psychological aspects of the game

III. SIMULATION RUN SUMMARY

A. Overview of Runs

For this milestone, we conducted several simulation runs to evaluate different aspects of the system. Each run was designed to test specific components or scenarios:

- 1) **Baseline Runs:** Initial simulations using simplified models and random event generation to establish baseline match dynamics and statistics
- 2) **LEM Model Runs:** Simulations using our trained Large Events Model to evaluate the quality of on-ball action predictions
- 3) **Movement Model Runs:** Tests focused on off-ball player movement and positioning using our transformer-based model
- 4) **Integrated Runs:** Full system simulations combining both models to evaluate overall match realism
- 5) **Tactical Variation Runs:** Simulations with different team formations and tactical styles to assess their impact on match outcomes
- 6) **Performance Benchmarking:** Runs designed to measure computational efficiency and identify bottlenecks

These runs collectively enabled us to assess the system's ability to generate realistic match scenarios, the accuracy of our prediction models, and the overall fidelity of the simulation relative to real UCL matches.

B. Simulation Run Table

The full simulation run table is provided in Table I. This table details the specific parameters, configurations, and outcomes for each simulation run. Key metrics include:

- Match statistics (possession, shots, goals)
- Event counts and distributions
- Model performance metrics
- Computational efficiency measures

C. Key Insights

Several significant insights emerged from our simulation runs:

- 1) The discrete-event approach effectively captures the flow of football matches, with realistic transitions between different phases of play
- 2) Player positioning is highly sensitive to tactical configurations, with formation choices significantly impacting passing networks and attacking patterns
- 3) The transformer-based off-ball movement model successfully replicates emergent team behaviors such as defensive compactness and attacking width
- 4) Environmental factors, though not explicitly modeled, can be approximated through adjustments to player movement and pass success probabilities

- 5) The probabilistic nature of event outcomes creates natural variability in match results, reflecting the inherent unpredictability of football

We also identified areas requiring improvement, particularly in modeling player decision-making in complex scenarios and better capturing the effects of fatigue and momentum. These insights will guide our future development efforts.

IV. DATA COLLECTION AND STORAGE

A. Data Collection Methodology

Our data collection approach encompasses multiple sources and methods to ensure comprehensive coverage of the factors influencing football match dynamics:

1) Player Attributes:

- We collected detailed player statistics from FBref using our custom web scraper (`webscraper.py`)
- Attributes include technical skills (passing, shooting), physical characteristics (pace, stamina), and positional tendencies
- Data was normalized to ensure consistency across different statistical categories

2) Team Tactics and Styles:

- Team formations, playing styles, and tactical preferences were collected
- Tactical dimensions include possession style, defensive approach, width, and attacking mindset
- Each tactical attribute was encoded to facilitate integration with our prediction models

3) Match Event Data:

- We leveraged event data from StatsBomb's open data repository
- Events include passes, shots, carries, tackles, and various other on-ball actions
- Each event contains spatial coordinates, timestamps, and outcome information

4) Simulation-Generated Data:

- Our simulations themselves generate significant data for analysis
- This includes detailed player movements, event sequences, and match outcomes
- This data serves both for analysis and for further training of our prediction models

Data collection was automated through Python scripts, with appropriate rate-limiting and error handling to ensure reliable acquisition. The web scraper implemented in `webscraper.py` systematically extracts data from structured tables on FBref, handling multi-level column headers and performing initial data cleaning.

B. Data Formats and Storage

The collected and generated data is organized in a structured format to facilitate analysis and model training:

1) CSV Files:

- Player data: `players.csv` - Contains player attributes and characteristics
- Team data: `teams.csv` - Basic team information
- Team styles: `team_styles.csv` - Tactical configurations and playing styles
- Events: Various event-related CSVs capturing match actions

2) SQLite Database:

- Core tables: `players`, `teams`, `team_styles`, `events`
- The database schema ensures referential integrity between related entities
- Indexing optimizes query performance for common analysis patterns

3) Model Storage:

- Trained models are saved using PyTorch's native serialization
- Model parameters, architectures, and preprocessing information are preserved
- This enables consistent loading of models for inference during simulation

4) Simulation Output:

- Event logs: Chronological records of all match events
- Position histories: Time-series data of player positions
- Match statistics: Aggregated metrics and performance indicators

The data storage approach prioritizes both accessibility for analysis and efficiency for model training. Regular backups ensure data integrity, while versioning enables tracking of how model performance evolves over time.

V. DATA ANALYSIS

A. Analysis Approach

Our analysis of simulation data employs both statistical methods and visual techniques to extract meaningful insights:

1) Event Distribution Analysis:

- We analyze the frequency and spatial distribution of different event types
- Comparisons with real match data help validate simulation fidelity
- Statistical tests assess whether simulated event distributions match empirical observations

2) Player Movement Analysis:

- Heat maps visualize player positioning and movement patterns
- Network analysis examines passing connections and team structures
- Comparison of player movement with and without the ball reveals tactical roles

3) Model Performance Evaluation:

- For the LEM, we evaluate prediction accuracy across different event types
- For the movement model, we assess the realism of predicted player positions

- Cross-validation ensures robust performance estimates

4) **Outcome Analysis:**

- Match results are analyzed across multiple simulation runs
- We examine how team attributes and tactical choices influence outcomes
- Statistical significance is assessed for key relationships

5) **Sensitivity Analysis:**

- Systematic variation of model parameters helps identify critical factors
- This reveals which aspects of the simulation most strongly influence results
- Helps prioritize future improvements based on impact

The analysis leverages Python's data science ecosystem, including Pandas for data manipulation, Matplotlib and Seaborn for visualization, and SciPy for statistical testing.

B. Results

The analysis of our simulation runs has revealed several noteworthy patterns and relationships:

1) **Event Distributions:**

- The distribution of event types (passes, shots, tackles) aligns reasonably well with observed patterns in real UCL matches
- Spatial patterns show expected concentrations in key areas (final third for shots, middle third for passes)
- Success rates for different actions correlate appropriately with player skill levels

2) **Player Movements:**

- The transformer-based movement model successfully captures team-level patterns such as defensive compactness and attacking width
- Player movement heatmaps show position-appropriate spatial coverage
- Distance metrics reveal realistic running patterns that vary by position and match context

3) **Team Tactics:**

- Different tactical configurations produce distinct patterns of play
- Possession-oriented teams show higher pass counts and wider spatial distribution
- Counter-attacking setups generate fewer but more direct attacking sequences

4) **Model Performance:**

- The LEM achieves reasonable accuracy in predicting event types and outcomes
- The movement model generates positionally sound off-ball behavior
- Integrated performance shows cohesive match flow with appropriate transitions between game states

C. Discussion

The results of our analysis provide significant insights into both the strengths of our current approach and areas for

improvement:

1) Strengths:

- The discrete-event simulation successfully captures the fundamental dynamics of football matches
- The integration of machine learning models enables realistic, data-driven behavior
- Player attributes meaningfully influence action outcomes, creating authentic performance variations
- Team tactical styles produce recognizable patterns of play

2) Areas for Improvement:

- Event sequencing sometimes lacks the organic flow seen in real matches
- Player decision-making in complex scenarios can appear mechanical
- The influence of match context (score, time remaining) on player behavior needs refinement
- Team-level tactical adaptations in response to match circumstances are limited

These findings suggest that while our core simulation approach is sound, enhancements to the ML models' contextual awareness and the implementation of more sophisticated tactical reasoning would significantly improve realism. Additionally, incorporating more diverse training data would help address some of the observed limitations in unusual match scenarios.

VI. CONCLUSION

A. Summary of Findings

Our simulation runs and data analysis have demonstrated that the integration of discrete-event simulation with machine learning models provides a promising foundation for realistic football match simulation. The system successfully captures many fundamental aspects of match dynamics, including:

- 1) The relationship between player attributes and action outcomes
- 2) The influence of team tactics on spatial patterns and game flow
- 3) The probabilistic nature of football events, creating natural variability
- 4) The complex interplay between on-ball actions and off-ball movements

The Large Events Model (LEM) effectively predicts plausible sequences of on-ball actions, while the transformer-based movement model generates coherent positional behavior. Together, these components create a simulation that, while still having limitations, captures many essential characteristics of real football matches.

B. Implications for the Project

These findings have several important implications for the project's direction:

- 1) The discrete-event approach is validated as an effective framework for football simulation
- 2) Machine learning models can successfully capture the complex decision-making processes in football

- 3) The integration of player attributes, team tactics, and match context creates emergent behaviors that resemble real match patterns
- 4) Additional refinement is needed in specific areas, but the core architecture is sound
- 5) The data generation capabilities of the simulation create opportunities for further model training and improvement

The current implementation provides a solid foundation for both analytical applications (understanding team performance factors) and predictive use cases (exploring tactical variations and match outcomes).

C. Next Steps

Based on our findings, we plan to pursue several key enhancements in the next phase:

1) Improved Contextual Awareness:

- Enhance models to better account for match state (score, time remaining)
- Implement more sophisticated tactical adaptation mechanisms
- Develop better models of psychological factors (momentum, pressure)

2) Enhanced Data Integration:

- Incorporate additional data sources for more comprehensive player characterization
- Develop better methods for translating tactical concepts into model parameters
- Implement continuous learning to refine models based on simulation outcomes

3) System Optimization:

- Improve computational efficiency to enable more extensive simulation runs
- Refine the event scheduling system for better performance
- Implement parallel processing for certain simulation components

4) Validation Framework:

- Develop more rigorous methods for comparing simulation outputs with real match data
- Create benchmarks for assessing simulation fidelity across multiple dimensions
- Establish quantitative metrics for evaluating model improvements

These next steps will address the identified limitations while building on the strengths of the current implementation, moving us closer to our goal of a highly realistic UEFA Champions League simulation.

VII. DOCUMENTATION UPDATES

A. Model Changes

Several significant updates have been made to the models since the previous milestone:

1) Large Events Model (LEM):

- Implemented a multi-head architecture with specialized components for event type, spatial coordinates, and outcome prediction
- Enhanced the feature extraction layers to better capture sequential patterns in match events
- Added temperature-based sampling for more natural event generation
- Implemented a simplified version for rapid prototyping (SimpleLEM)

2) Off-Ball Movement Predictor:

- Adopted a transformer-based architecture to better model player interactions
- Incorporated attention mechanisms to capture dependencies between player positions
- Added specialized embeddings for player attributes, team tactics, and spatial coordinates
- Implemented position-relative encoding to improve spatial awareness

3) Integration Framework:

- Developed a unified simulation engine that orchestrates both models
- Implemented event scheduling and processing mechanisms
- Created interfaces between models and the simulation state
- Added visualization capabilities for debugging and analysis

These changes have significantly enhanced the models' ability to generate realistic football behavior and improved the cohesion of the overall simulation.

B. Tools and Libraries

The project now leverages several additional tools and libraries:

1) PyTorch (v1.10+):

- Core framework for all machine learning models
- Used for both model training and inference during simulation

2) NumPy (v1.20+):

- Fundamental array operations and numerical computing
- Supports spatial calculations and statistical analysis

3) Pandas (v1.3+):

- Data manipulation and analysis
- Used for processing event data and aggregating simulation results

4) Matplotlib (v3.5+):

- Visualization of simulation results
- Creation of pitch plots, heatmaps, and event distributions

5) BeautifulSoup (v4.9+):

- Web scraping for data collection
- Used to extract player and team statistics from FBref

6) **SQLite3:**

- Lightweight database for storing player, team, and event data
- Provides efficient query capabilities for data analysis

7) **Scikit-learn** (v1.0+):

- Data preprocessing and encoding
- Feature engineering for model inputs

These tools collectively enable efficient data processing, model development, simulation execution, and results analysis.

C. Simulation Plan Table Updates

The simulation plan has been refined based on our current findings:

1) **Additional Scenarios:**

- Added specific tactical matchup scenarios (e.g., high-press vs. low-block)
- Included fatigue impact assessment scenarios
- Added player substitution effect analysis

2) **Expanded Metrics:**

- Added expected goals (xG) as a key validation metric
- Incorporated player movement distance tracking
- Added passing network analysis metrics

3) **Revised Timelines:**

- Extended model training phase to accommodate more extensive data collection
- Added dedicated optimization phase to improve computational efficiency
- Increased the number of validation runs to ensure statistical significance

4) **New Model Benchmarks:**

- Established comparison standards against statistical baselines
- Added real match replay comparisons for visual validation
- Created quantitative metrics for assessing simulation realism

These updates reflect our evolving understanding of the project requirements and the insights gained from the current milestone's simulation runs.

VIII. REFLECTION AND NEXT STEPS

A. Challenges Faced

During this milestone, we encountered several significant challenges:

1) **Data Integration Complexity:**

- Merging disparate data sources with different formats and scales
- Ensuring consistency between player attributes, team tactics, and event data
- Handling missing or incomplete data in certain categories

2) **Model Training Limitations:**

- Limited availability of comprehensive training data for certain event types
- Balancing model complexity with computational efficiency
- Achieving appropriate generalization across different match contexts

3) **Simulation Stability:**

- Ensuring consistent behavior across long simulation runs
- Handling edge cases and unusual event sequences
- Maintaining physical realism in player movements and interactions

4) **Computational Constraints:**

- Managing memory usage during extensive simulation runs
- Optimizing performance for real-time or near-real-time operation
- Balancing fidelity with processing requirements

5) **Validation Methodology:**

- Defining appropriate metrics for assessing simulation realism
- Obtaining suitable benchmark data for comparison
- Quantifying subjective aspects of match dynamics

These challenges required thoughtful problem-solving and sometimes necessitated compromises between different project objectives.

B. Solutions Implemented

To address these challenges, we implemented several effective solutions:

1) **For Data Integration:**

- Developed a unified data preprocessing pipeline with standardized scaling
- Created mapping functions to relate different entity identifiers across sources
- Implemented imputation strategies for handling missing data

2) **For Model Training:**

- Employed data augmentation techniques to expand training datasets
- Implemented transfer learning for feature extraction components
- Used gradient accumulation to enable larger effective batch sizes

3) **For Simulation Stability:**

- Added comprehensive state validation checks throughout the simulation
- Implemented graceful fallback mechanisms for edge cases
- Created detailed logging for diagnosing unusual behaviors

4) **For Computational Efficiency:**

- Optimized tensor operations for inference speed

- Implemented batched processing where appropriate
- Used selective model activation based on context importance

5) For Validation:

- Developed multi-faceted evaluation criteria combining statistical and visual assessment
- Created synthetic baseline comparisons for contexts lacking real data
- Implemented progressive validation at multiple levels of abstraction

These solutions significantly improved the overall performance and reliability of the simulation system while providing a foundation for future enhancements.

C. Future Plans

Building on the current milestone, our plans for future development include:

1) Enhanced Data Collection:

- Expand data sources to include additional leagues and seasons
- Incorporate more detailed physical tracking data where available
- Develop methods for extracting tactical information from video sources

2) Model Improvements:

- Implement more sophisticated sequence modeling for the LEM
- Enhance the transformer architecture for better long-range dependencies
- Develop specialized models for set-piece scenarios

3) Simulation Expansion:

- Extend the simulation to model full tournaments, including group stages and knockout rounds
- Implement inter-match effects such as fatigue accumulation and suspension management
- Add team selection and substitution strategies

4) Interactive Capabilities:

- Develop interfaces for real-time intervention in simulations
- Create visualization tools for tactical analysis
- Implement scenario testing functionality for what-if analysis

5) Validation Enhancements:

- Establish collaboration with domain experts for qualitative assessment
- Implement automated comparison with real match data
- Develop benchmarks for specific tactical scenarios

These plans will guide our work in the upcoming milestones as we continue to refine and enhance the UEFA Champions League simulation system.

TABLE I
SIMULATION RUN TABLE

Run	Goal	Config	Expected	Actual	Notes
1	Baseline test	Barcelona vs. Bayern (4-3-3 vs. 4-2-3-1)	Basic event generation	2-1, 120 events	Event skew toward passes
2	Test LEM	Liverpool vs. Real Madrid (4-3-3)	Realistic sequencing	0-0, 156 events	Improved transitions
3	Test movement model	Man City vs. PSG (4-3-3 vs. 4-2-3-1)	Realistic positions	2-2, formation visible	Good defense model
4	Evaluate tactics	Possession vs. Counter-attack	Distinct patterns	1-3, counter effective	Clear statistical differences
5	Benchmark performance	Full sim with viz	Real-time performance	20% slower	Position update bottlenecks
6	Test set-pieces	Corners and free kicks	Realistic execution	Limited variety	Needs enhancement
7	Extra time	Knockout match	Fatigue effects	Visible decrease in performance	Promising but needs calibration
8	Substitution effects	Multiple subs	Fresh player impact	Higher activity	Effect smaller than expected

IX. REFERENCES

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